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The Coordination Substrate: A Detection Engineering Analysis of the September 2026 Unit 42 Agentic Intrusion, and Why Provider-Egress Detection Fails Against Borrowed Inference

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Fourth in the consolidated detection engineering series. Direct successor to *Agentic Intrusion* (concept DOI 10.5281/zenodo.22033405), whose Tier 1 claims this paper tests and, in one respect, narrows.

Scope and provenance

Built from Unit 42’s public account of an AI-assisted intrusion, published 2 September 2026, and re-read against Dream’s account of the July 2026 campaign against Taiwanese government and energy targets. Contemporaneous reporting is used only for facts the primary accounts do not carry, and is labelled where it appears.

We had no victim telemetry, no samples, no forensic artifacts, and no contact with either affected organisation. Nothing was reproduced and nothing was detonated. No rule in the companion package has been executed against telemetry from either incident, because none is public.

ATT&CK and ATLAS mappings are our inference from described behaviour. Neither source supplied one.

The correction this paper carries. *Agentic Intrusion* shipped provider-egress detection as Tier 1, on the argument that an agent must reach an inference endpoint on every planning step. That argument holds. Section 5 shows a phase of the Unit 42 intrusion in which it produces silence, for a reason we anticipated in outline and mis-specified in mechanism.

1. The defender question nobody is answering

Thirteen days after we published a tempo model for agent-driven intrusion, an independent investigator published hunt guidance naming the same signals, and described a phase of the same class of intrusion that defeats the other half of what we shipped.

Two questions follow, and the coverage of this incident answers neither:

When two teams independently propose the same detection primitives and neither has measured them, what has actually been established?

What do multi-agent intrusions leave behind that survives the operator changing model, provider, framework and infrastructure?

The first is a question about evidence, and getting it wrong inflates confidence across the whole discipline. The second has an answer that is visible in both published corpora and has not, as far as we can find, been named.

2. The incident

Unit 42 investigated an intrusion at an enterprise running cloud infrastructure and its own AI services. The motive was ransom. No actor attribution was offered and we offer none.

Roughly ten hours of operational activity, spanning more than fifty MITRE ATT&CK techniques, against an estimated two weeks for an equivalent human operation.

The described chain:

Stage	What happened
Infiltration and mapping	Breach of a publicly reachable web service, then an automated reconnaissance agent mapping the microservice estate
Secrets harvesting	Sub-agents extracting hardcoded tokens and passwords from code repositories
Privilege takeover	Compromise of the secrets management system, yielding root administrative credentials
Pipeline exploitation	CI/CD hijack, with an attempted Terraform backdoor blocked by branch protection
AI infrastructure hijack	Stolen cloud keys used to repurpose the victim's own AI endpoints as post-compromise infrastructure

Unit 42 separates what it observed from what it assessed, and the distinction is load-bearing for sections 3 and 4, so it is preserved here.

Observed. LLM calls to multiple frontier AI agents in parallel. Structured Markdown files passing information between agents and sessions. Python caches. Paired asset folders. Markdown formatted outputs. An eighty-page technical audit of the victim’s security findings, left behind.

Assessed. Custom scripts judged with high confidence to be AI-generated, on the basis of their UI elements.

Recommended for hunting, and not presented as measured. Bursty API requests, rapid 401 to 200 HTTP state shifts, parallel authentications, and sudden model usage from unexpected identities.

3. Convergence is not confirmation

Agentic Intrusion argued that agent-driven activity produces a bimodal rhythm: machine-speed action bursts separated by inference-latency pauses, with technique change following failure. It listed as its first falsification condition that agent-driven and script-driven intrusions might prove indistinguishable on that joint distribution in real telemetry.

Unit 42’s hunt guidance names bursty API requests, parallel authentications from unexpected identities, and sudden model usage. That is recognisably the same shape, arrived at independently, from a different incident, with a different victim and a different motive.

It is tempting to read that as corroboration. It is not, and the reason is worth stating precisely because the error is about to become common.

Unit 42 published no interval data, no request rates, no timestamps and no latency measurements. Their list is guidance about where to look. Ours is a model reasoned from published inference latencies. Neither is a measurement. Two teams reasoning from the same structural facts about how inference-driven planning works will reach the same primitives, and their agreement is evidence that the reasoning is unremarkable, not that the detection fires on real activity at a usable rate.

What the convergence does establish is modest and real: the inference from architecture to rhythm is obvious enough that two groups made it independently, which slightly raises the prior that it is correct. What it does not establish is any of the things a practitioner needs, namely the threshold, the base rate, or the false-positive cost in a production estate.

The compounding hazard. A third party citing both sources will count two independent confirmations of a tempo signature. There are zero. If this paper contributes nothing else, it should contribute that sentence, because the failure mode is the discipline convincing itself that a plausible model is a validated one by circulating it.

Falsification condition 1 from *Agentic Intrusion* is therefore **still open**. We would rather say that than accept a corroboration we did not earn.

4. The coordination substrate

This section is the contribution.

4.1 The overlap between the two corpora is not a technique

Set the two published corpora side by side and remove everything that is specific to one.

	Taiwan, July 2026 (Dream)	Enterprise, ~September 2026 (Unit 42)
Operator	unattributed, Chinese-language artifacts	unattributed
Motive	espionage	ransom
Victim	government, then a nuclear safety regulator and energy firms	enterprise with cloud and AI services
Duration	four days, twelve waves	under ten hours
Frameworks	Hermes and OpenClaw, open source	unnamed frontier agents plus agentic frameworks
Shared indicator	none	none
Shared artifact class	1,395 files in a 160 MB workspace; .hermes and .openclaw workspace directories; sub-agents lettered A through Q; structured after-action reporting between waves; recorded posterior scoring of findings	structured Markdown files passing information between agents and sessions; paired asset folders; Python caches; an eighty-page technical audit

Nothing in the top half of that table matches. The bottom row does, and it matches closely: in both incidents, the orchestration produced a body of structured, cross-referencing state files as a by-product of running.

4.2 Why it is a property of the architecture rather than a choice

A single agent needs no coordination artifacts. Its state is its context window.

Several agents do, and the reason is mechanical:

- **Context windows are finite and metered.** Multi-agent designs exist to partition work that will not fit, or will not fit affordably, into one window. Partitioning the work partitions the state.
- **Sub-agents are separate processes, frequently separate sessions.** The only channel between two processes that survives the end of one of them is storage.
- **Planners that re-rank need the prior evidence.** Re-sending the full history on every call reintroduces exactly the cost the architecture was adopted to avoid, so the evidence is written down and referenced.

Every one of those pressures points the same way. An operator can change model, provider, framework, infrastructure and technique for free, and still has to put the shared state somewhere the next agent can read it.

That is the same form of argument *Agentic Intrusion* made for tempo. The difference, and the reason this paper leads with the substrate rather than the rhythm, is that **the substrate has been observed twice and the rhythm has been measured zero times**.

4.3 What the signal actually is, stated narrowly

It is not “Markdown files” and it is not “`__pycache__`”. Both are ubiquitous and a rule keyed on either is noise.

The signal is the **conjunction**: structured state files, created at machine cadence, by a non-interactive process, on a host with no developer workflow, cross-referencing each other, in the same window as authentication or reconnaissance activity. Any one element is ordinary. The combination describes a program writing notes to itself faster than a person could read them, on a machine where nobody writes notes.

4.4 The vantage problem applies, and halves the reach

Asserted Egress (concept DOI 10.5281/zenodo.22314504) named the vantage problem: an incident can be richly indicated on one side of a boundary and structurally invisible on the other. It applies here and it constrains this contribution more than we would like.

In the Unit 42 case the agents ran **inside the victim estate**, so the substrate landed on victim hosts and is a defender-visible artifact.

In the Taiwan case the workspace was on the **operator’s** infrastructure. It became available to researchers because the operator exposed a 160 MB archive to the internet, not because a defender found it. A Taiwanese defender saw authentication activity and nothing else.

So the substrate is detectable exactly when the operator runs agents on your machines. When they run them on their own and reach you through ordinary protocols, you see none of it. Roughly half the published corpus is out of reach of every rule in section 6’s Tier 1 first entry, and any pack that does not say so is selling coverage it does not have.

5. Borrowed inference

The second contribution, and it is a correction to our own August package.

5.1 The blind spot

Agentic Intrusion shipped two provider-egress rules as its highest-value, lowest-cost control: a server or service context resolving a public model provider, and inference egress from a restricted network zone. The reasoning was that the agent must call an inference endpoint on every planning step, and that this is a structural consequence of the method rather than an operator choice.

Unit 42’s fifth stage is the case where that produces nothing. Stolen cloud keys were used to repurpose **the victim’s own AI endpoints** as post-compromise infrastructure. In that configuration:

- the destination is inside the victim’s own tenancy, not a public provider;
- it is allowlisted by construction, because the victim runs AI services deliberately;
- the traffic need not cross the perimeter the proxy rule watches;
- the credential is valid, so nothing fails.

Both August rules fail to silence. Not to a false negative that a tuned threshold recovers: to silence, because the destination they key on is never contacted.

5.2 What we got right, and what we got wrong

The August package’s limitations section says that self-hosted inference “defeats the egress rules completely”, and predicts this becoming the common case as open-weight model quality improves.

The failure mode was right. The mechanism was wrong, and the timeline was wrong. It did not arrive through operators hosting their own models. It arrived through operators **using the victim’s**, and it arrived in the next disclosed incident rather than in some later era of open weights.

The general lesson is worth more than the specific rule: a detection keyed on a **destination** fails whenever the attacker acquires a legitimate instance of that destination. Keying on **identity** does not have that failure mode.

5.3 The structural cause, named

Borrowed inference: an attacker who compromises an estate that operates AI workloads inherits its inference capability along with its data and its compute, and that inference is indistinguishable from the estate’s own by every network-layer control the estate owns.

It is the same shape as stolen cloud credentials used for cryptomining, with one difference that matters for detection: mining is an anomalous workload on general compute, while borrowed inference is the **expected** workload on the exact service built to serve it. It looks like the product working.

It generalises past language models to any metered capability an estate exposes to itself: managed inference, transcription, translation, OCR, rendering farms. Wherever an organisation has paid to make a capability easy to call from inside, an intruder inside gets it at the same price.

6. Detection, ranked by durability

Ranked by what survives an operator changing things, not by ease of deployment. Full rules, with false positives and tuning notes, are in the companion package.

Tier 1: survives model, provider, framework and infrastructure change

1. Model endpoint invoked by a principal with no invocation history. The borrowed-inference control, and the highest-value new rule in this pack. Cloud model-invocation logs carry the calling principal. The question is not whether a host talked to a model provider but whether **this identity has ever invoked this endpoint before**, and whether its rate changed by an order of magnitude. A service principal whose job is document processing calling an inference endpoint for the first time, at volume, is the whole finding. Immune to provider choice, framework choice and network position, because it is asked in the control plane.

2. Coordination artifacts created at machine cadence by a non-interactive process. Section 4.3’s conjunction, expressed against file-creation telemetry. A **hunting** rule, not an alert, and vantage-limited per section 4.4. Its value is that it fires during the operation rather than at staging, and that it is the one artifact class observed in both published corpora.

Tier 2: durable, requires baselining before it is enabled

3. Failed-to-successful authentication transition rate, joined to session parallelism. Unit 42 names the 401 to 200 shift. On its own it is a credential-spray signal that any brute-force detection already carries. What separates an agent swarm from a spray tool is the **join**: multiple concurrent sessions on the same credential performing *different* actions. Spray tools repeat one action widely; agents divide work. The parallelism join is ours; the transition rate is Unit 42’s.

4. Step change in model invocation volume per principal. Cheap, and it catches borrowed inference where rule 1’s history baseline is unavailable because the principal is new.

Tier 3: useful, decays quickly

5. AI-generated tooling heuristics. Unit 42 assessed scripts as AI-generated from their UI elements. We are not building alerting on that and we advise against it. Stylistic tells are the least durable signal in this paper: they are free to remove, they are already being removed, and a false positive lands on your own developers. Recorded because it appears in the source, ranked last because it should be.

7. What this class makes undetectable

- **The substrate is invisible when the agents run elsewhere.** Section 4.4. Taiwan is that case.
- **Both corpora exist because of operator error.** A 160 MB exposed archive and an eighty-page audit left on a victim host. Neither is a control. Do not build a program on the assumption that the next operator is as untidy.
- **Borrowed inference is invisible without model-invocation logging,** which is off by default or absent in many tenancies, and is frequently retained for days rather than months.
- **Tempo remains unmeasured,** by us and by Unit 42. Section 3.
- **The bypass, wherever one occurs, happens inside a prompt.** Unchanged from the August paper: unless you operate the gateway, that exchange is not yours to see.
- **Ten hours is inside most detection and response cycles.** A control that fires correctly on day two is a post-mortem input, not a defence.

8. What actually found it

Fourth consecutive paper in this series, and the answer is again not the defender’s detection stack.

Incident	What found it
Hugging Face, July 2026	the victim, and a competitor’s public disclosure
Taiwan, July 2026	researchers finding an exposed operator archive

Incident	What found it
Wiki collusion, May to July 2026 This incident, September 2026	a volunteer moderator reading an edit log incident response after the fact, materially assisted by an audit the attacker left behind

Four incidents, four detections, none of them a rule that fired. This section is short and unflattering on purpose, and it is the most consistent finding in the series.

9. Credit

Unit 42 published hunt guidance rather than a capability advertisement, and separated what they observed from what they assessed with confidence. That separation is the entire basis of section 3. Had they presented their hunt list as measurement, we would have read it as corroboration and been wrong. Careful epistemic labelling in a vendor writeup is rarer than it should be and it did real work here.

The same applies to Dream, whose account qualified its own headline: near-autonomous rather than autonomous, with an explicit note that building the system took more than running a model.

10. Limitations

No telemetry, no samples, no victim contact. Every technical claim traces to a published account.

Unit 42's writeup is a summary. A fuller technical report may contradict specifics here. This paper is current to 8 September 2026 and should be re-read against any subsequent publication.

n equals two, and both are selected. The two corpora are what two vendors chose to publish. Multi-agent intrusions that were never noticed, or noticed and not disclosed, are unmeasurable by construction, and there is no reason to think the disclosed ones are representative.

The substrate argument is architectural, not empirical. Section 4.2 reasons from how multi-agent systems must work. It predicts the artifact class rather than being derived from a survey of it. Section 11 states what would break it.

Mappings are inferred. Neither source supplied ATT&CK or ATLAS mappings.

Nothing was executed against real telemetry. Sigma is schema-validated, KQL and SPL are authored and not run. The companion package states this per format.

11. What would falsify this

1. **If coordination state moves in-band** - agents exchanging state through a command channel, a provider's own session storage, or a shared context service rather than the filesystem - the substrate stops touching disk and section 4 describes a transitional artifact. This costs the operator very little and we consider it the most likely of the four.

2. **If structured state files plus interpreter caches prove indistinguishable from ordinary developer and CI activity** at production scale, Tier 1 rule 2 is noise regardless of the architecture argument.
 3. **If model-invocation logging is unavailable or unaffordable in most estates**, Tier 1 rule 1 is uninstrumentable even if it is correct, which would leave this paper with no deployable Tier 1 at all.
 4. **If a third corpus shows multi-agent intrusion with no persistent shared state**, the pressures in section 4.2 are not as binding as claimed and the substrate is an implementation habit rather than a property.
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12. What we are not claiming

Not that Unit 42 confirmed our tempo model. They tested nothing and neither did we. Section 3 exists to prevent that reading.

Not attribution, for either incident. Unit 42 offered none. Dream declined to name a group.

Not that these are the same operator. Different motive, different victim class, different duration, no shared indicator. The shared artifact class is evidence about architecture, not about people.

Not that running AI services is unsafe. Borrowed inference is a consequence of holding a valuable metered capability, in the same way that holding compute invites mining. The mitigation is per-principal authorization and invocation logging, not the absence of the service.

Not that the August package was wrong. Its egress rules work against the case they describe, and that case is the majority of what has been reported. They have a blind spot, now named, and the correct response is an additional identity-keyed control rather than a retraction.

Not that ten hours is the floor. It is the fastest published, which is not the same thing.

13. AI disclosure

Prepared with AI assistance (Anthropic Claude) inside the author’s research harness. The coordination substrate synthesis, the borrowed inference naming, the convergence argument in section 3, the durability rankings, the falsification criteria, and sections 7 through 12 were authored and reviewed by the named human author, who is solely responsible for the claims. The limitations and non-claims sections were written before the rules rather than after, because AI-assisted detection content tends to overstate coverage, and section 3 exists specifically because the convenient reading of the evidence was the wrong one.

References

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- Saluca Labs. *Detection Without Indicators: Agent-Originated Intrusion*. Concept DOI 10.5281/zenodo.21770780.
- Companion detection content: <https://github.com/saluca-labs/detection-content>, Apache-2.0.

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